

4Suboi Zenabu first version.docx

by Zenabu Suboi

Submission date: 30-Mar-2025 08:21PM (UTC+0200)

Submission ID: 2629631301

File name: 4Suboi_Zenabu_first_version.docx (24.1K)

Word count: 689

Character count: 4461

Scientific Writing – Assignment 4 – Zenabu Suboi

Research Topic: Understanding the key drivers of impact of malaria interventions in Ghana

Possible introductory sentences

1. General statement

Malaria, an infectious disease caused by the Plasmodium falciparum parasite kills over half a million people every year.

2. Statistics

In 2023, malaria killed 597'000 people with 263 million cases globally.

3. General statement and statistics

Malaria is a threat to human life worldwide with 263 million cases and 597'000 deaths in 2023.

Introduction

Malaria is a deadly disease with global estimates of 263 million cases and 597'000 deaths in 2023. Sub-Saharan Africa accounted for 94% of this global burden with 95% deaths (1). With over 5 million cases in a population of 33 million in Ghana, the disease imposes a substantial burden on the population, particularly among pregnant women and under five year-olds (2,3).

Since the colonial era, Ghana has commitment to several malaria control initiatives both national and international, which has resulted in a steady progress in its fight against malaria. Due to this consistent progress, the national malaria control program (NMCP) of Ghana, in 2022, shifted its focus from the control of malaria to its elimination, becoming the national malaria elimination program (NMEP). This shift in focus necessitated the development of the national malaria elimination strategic plan (NMESP 2024-2028) to match its new direction, while the national malaria strategic plan (NMSP 2021-2025) is still in operation. One of the three goals of the NMESP is to eliminate malaria in 21 districts of the country with very low malaria burden or transmission (districts with 1-49 cases per 1000 population or <1% parasite prevalence) by 2028. Toward achieving this goal, the NMEP has scheduled intervention packages to be deployed at the district level, based on malaria risk stratification (4).

Mathematical models of malaria transmission have been extensively used to tailor malaria interventions at the subnational level to select appropriate intervention packages for specific risk areas (5,6). Determining the relative contribution of each parameter of these models to the model outputs allows to identify the key drivers of disease dynamics (7). Geographic-specific knowledge of the key drivers of impact of interventions on local malaria transmission and burden is crucial in guiding the planning of malaria control programs. Such knowledge will ensure the optimal allocation of limited resources, particularly in this era where the future of the US President's Malaria Initiative (PMI) funding is uncertain (7). Global Sensitivity analysis of malaria transmission models can help in determining the relative importance of individual interventions to malaria prevalence within a specific geographical area (8,9). However, to the best of our knowledge, such a comprehensive analysis to identify these key drivers of impact of interventions in the

districts of Ghana has not yet been conducted. In this study, we perform a global sensitivity analysis using the Sobol' approach to determine the relative importance of model parameters such as intervention coverage, efficacy and vector composition to malaria prevalence at the district level.

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